Comparing Classification Models on the Breast Cancer Wisconsin Diagnostic Dataset

DA7067 Computational Mathematics

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Dataset

I used the Breast Cancer Wisconsin (Diagnostic) dataset from the UCI Machine Learning Repository, loaded from the raw file `wdbc.data`. It contains 569 samples and 30 numerical features after removing the patient ID. The features describe cell characteristics from digitised breast mass images, measured as mean, standard error, and worst values. The label is diagnosis: 212 malignant (M) and 357 benign (B) cases, giving a class distribution of approximately 37% and 63%, respectively.

Question

I am curious whether the diagnosis can be predicted from routine cell-nucleus measurements alone, because fine-needle aspiration is far cheaper and faster than a surgical biopsy. This leads to two sub-questions:

1. How well can a model predict malignant vs. benign?
2. How does the complexity of the kNN model (its k) affect the result?

Frame the problem

This is a **supervised binary classification** task with two classes. The **features** are the 30 numeric cell-nucleus measurements; the **label** is the diagnosis (malignant or benign). It is supervised because every training sample comes with its known label, so the model can learn a mapping $f : \mathbb{R}^{30} \to \{\mathrm{M}, \mathrm{B}\}$ from labelled examples.

It is a **machine-learning problem rather than a classical-programming one** because no explicit rule relates the 30 measurements to the diagnosis. Even experienced pathologists may find it difficult to define an exact set of rules for making a diagnosis, because it depends on subtle patterns and relationships between several features. Instead of manually writing these rules, the model learns from examples with known diagnoses and uses what it has learned to make predictions for new patients. This is what makes supervised learning different from traditional programming: the model learns the rules from data rather than having them explicitly programmed.

Baseline and two models

I split the data 80/20 into a **training set** of 455 samples (285 `B`, 170 `M`) and a **test set** of 114 samples (72 `B`, 42 `M`), using **stratified sampling** so the ∼63/37 benign/malignant ratio is preserved in both parts. `random_state=42` makes the split reproducible.

Three classifiers were compared: a baseline classifier that always predicts the most frequent training class, logistic regression, and k-nearest neighbours (kNN).

The baseline doesn't use the features at all - it just predicts the most common class, so I evaluated it before standardising. For logistic regression and kNN I fitted `StandardScaler` on the training features only and then applied it to both training and test sets, so no test information leaks into the scaling. Standardising matters most for kNN, since its predictions rely on distances between points. Without it, a feature like *area* (range in the hundreds) would dominate one like *smoothness* (range ≈ 0.05–0.15).

Models

Three models were compared:

1. **Baseline:** `DummyClassifier(strategy="most_frequent")`. Predicts the most common class (benign) for every input, ignoring the features.
2. **Logistic regression:** a linear classifier (`max_iter=10000`).
3. **kNN with $k = 5$:** a local, non-parametric classifier.

Table 1 summarizes test performance.

Table 1: Model performance on the test set.

Model	Test accuracy	Test error
Baseline (most frequent class)	63.16%	36.84%
Logistic regression	96.49%	3.51%
kNN ($k = 5$)	95.61%	4.39%

Table 2 and Table 3 shows the confusion matrices and classification report metrics for the Baseline and Two models.

Table 2: Confusion matrices for the baseline, logistic regression, and kNN ($k = 5$).

Model	Actual class	Predicted B	Predicted M
Baseline	B	72	0
Baseline	M	42	0
Logistic regression	B	71	1
Logistic regression	M	3	39
kNN ($k = 5$)	B	71	1
kNN ($k = 5$)	M	4	38

Table 3: Classification report metrics for the baseline, logistic regression, and kNN ($k = 5$).

Model	Class	Precision	Recall	F1-score
Baseline	B	0.63	1.00	0.77
Baseline	M	0.00	0.00	0.00
Logistic regression	B	0.96	0.99	0.97
Logistic regression	M	0.97	0.93	0.95
kNN ($k = 5$)	B	0.95	0.99	0.97
kNN ($k = 5$)	M	0.97	0.90	0.94

As shown in Table 1, logistic regression performed best overall, achieving a test accuracy of 96.49% and a test error of 3.51%. The kNN model ($k = 5$) achieved a slightly lower accuracy of 95.61% and a test error of 4.39%. In comparison, the baseline model achieved only 63.16% accuracy because it always predicted the most frequent class, benign (B).

Table 2 provides further insight into these results through the confusion matrices. Logistic regression correctly classified 71 benign cases and 39 malignant cases, misclassifying only four cases in total. The kNN model correctly classified 71 benign cases and 38 malignant cases, with five misclassifications. The baseline correctly classified all 72 benign cases but failed to identify any of the 42 malignant cases.

The classification metrics in Table 3 support this comparison. For the malignant (M) class, logistic regression achieved a precision of 0.97, a recall of 0.93, and an F1-score of 0.95. The kNN model achieved the same precision of 0.97 but a lower recall of 0.90 and an F1-score of 0.94. This indicates that logistic regression identified more malignant cases correctly. The baseline had a recall of 0.00 for the malignant class, demonstrating that it was ineffective at detecting malignant tumours.

Overall, logistic regression was the best-performing method in this experiment because it achieved the highest test accuracy, the lowest test error, and the highest recall and F1-score for the malignant class. One possible explanation is that logistic regression learned an effective relationship between the numerical features and the diagnosis, while kNN relied on distances between neighbouring observations. However, the difference between the two models was small, and these results are based on a single train-test split. Further evaluation would be needed to determine whether logistic regression consistently performs better on unseen data.

Bias-variance trade-off

I applied kNN for $k = 1, 2, \ldots, 50$ and plotted both the training error (obtained by letting the model predict the training set itself) and the test error against k (Figure 1).

Where does the model overfit? At $k = 1$ **and** $k = 2$. At $k = 1$ the training error is exactly **0.000**, so the model predicts every training label perfectly while the test error is at ~ 0.070 and peaks at $k = 2$ (~ 0.088). This high variance shows that the model has memorised the training data, including its noise, and generalises poorly.

Where does it underfit? For **large** k **(roughly** $k \geq 20$**).** Training error rises to ~ 0.046 while test error plateaus at ~ 0.053, so the two curves converge. The model gets about 5% error on both training and test sets - almost the same number. When the two errors are equal and both are relatively high, it means the model is too simple to capture the real pattern in the data. This is underfitting.

Which k **would I choose, and why?** I would choose $k = 5$. It sits at the bottom of the U-shape in the test-error curve, the point where the model is flexible enough to capture the real pattern and smooth enough to ignore noise.

Choosing a smaller k would overfit and a larger k would underfit. The test error at $k = 5$ is **0.044**, the minimum, tied with $k = 7$. Between the two, I take the smaller value ($k = 5$) because when two models perform equally well, the simpler one is preferable. Strictly, k should be tuned on a separate validation set or via cross-validation rather than on the test set; here I followed the assignment's instruction to plot test error directly, and flag this as a limitation.

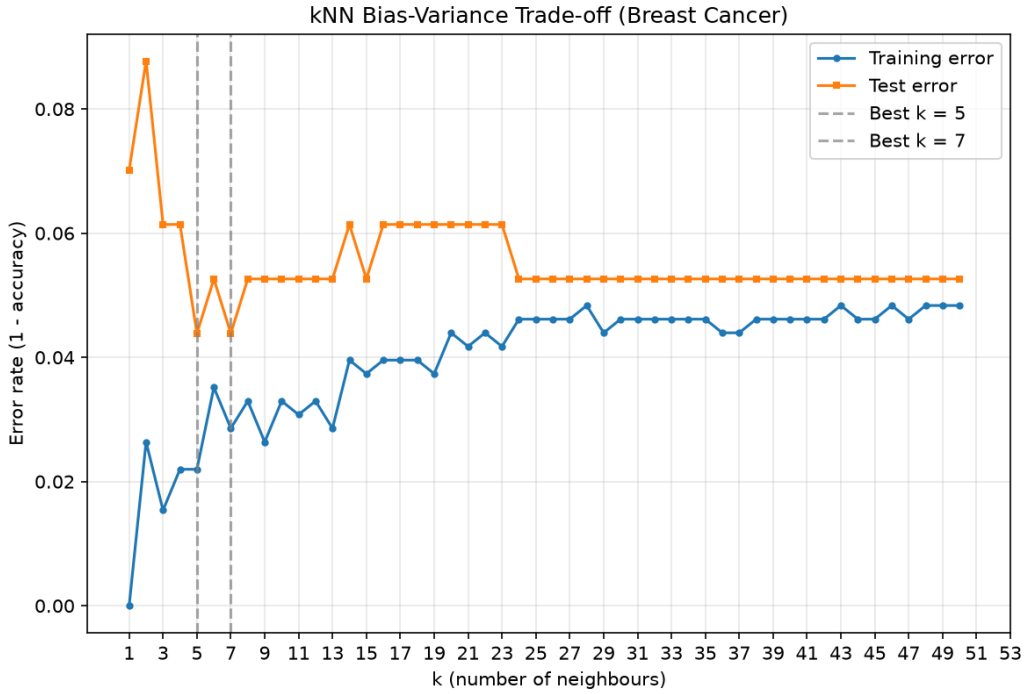


Figure 1: Training and test error for kNN over $k = 1, \ldots, 50$.

Reflection

What I would trust the model to do.

I would trust this model as a screening aid, to provide a preliminary prediction of whether a tumour is benign or malignant based on its numerical features. It can flag likely malignant cases for a clinician to review. On data drawn from the same distribution, using a similar measurement protocol and population, it achieved approximately $\sim 96\%$ accuracy and correctly identified 90-93% of malignant cases. It is fast

and consistent, and unlike a human, it does not get tired. However, these results come from a single test split and do not guarantee the same performance in clinical practice. Therefore, I would use the model to support clinical assessment, not replace a clinician or make diagnoses independently.

What I would not trust it with.

I would not trust the model to make medical diagnoses independently. On the test set, logistic regression missed 3 of the 42 malignant cases, while kNN missed 4. In a clinical setting, a missed malignancy can be far more serious than a false alarm. Any real-world use would require a larger, more diverse dataset and external validation.

I would also be cautious about applying the model to patients from different hospitals, scanners, or demographic groups, since differences in the data distribution could reduce its performance. Finally, I would not interpret the model's coefficients as evidence of causation. For example, a larger radius may be associated with malignancy, but this does not mean it causes malignancy. The model identifies patterns in the data rather than explaining the underlying biological mechanisms.

What surprised me.

Four things stood out.

First, I was surprised that logistic regression performed so well and even slightly better than kNN, achieving 96.49% accuracy compared with 95.61% for kNN ($k = 5$). For the malignant class, logistic regression achieved 0.97 precision and 0.93 recall, while kNN achieved 0.97 precision and 0.90 recall. This means logistic regression identified more malignant cases, although both models performed well. I expected the more flexible model to do better.

Second, high accuracy alone does not tell the whole story; recall is also important when identifying malignant cases.

Third, the **training error did not rise smoothly with** k. In theory it should increase as k grows, because a larger neighbourhood means a less flexible fit on the training data. In my results it went from 0.000 at $k = 1$ to **0.026 at** $k = 2$, then *dropped* to **0.015 at** $k = 3$ before rising again. The possible explanation is that an even k can produce tied votes in the nearest-neighbour vote. For example, one benign and one malignant neighbour at $k = 2$ and the tie is broken arbitrarily. With $k = 3$ ties are much rarer, so the training error falls. It's a small effect, but I only noticed it by looking carefully at the numbers rather than trusting the shape I expected.

Finally, how dramatically $k = 1$ fails. The model with the best training performance (training error 0.000) had the worst test performance. The gap between 0.000 and ~ 0.070 is the bias-variance trade-off made visible in a single number.

Conclusion

Both models predict the diagnosis well on this dataset: logistic regression reached **96.49%** accuracy and kNN ($k = 5$) **95.61%**, versus **63.16%** for a baseline that just predicts the majority class. Logistic regression missed 3 malignant cases, kNN missed 4 — good enough for a screening aid, not for a diagnosis.

Model complexity affects the result strongly, and in the way the bias-variance trade-off predicts. At $k = 1$–2 the model overfits (training error 0.000, test error up to 0.088); at $k \geq 20$ it underfits (both errors converge around 0.05). The best balance is at $k = 5$**, tied with** $k = 7$, giving the lowest test error of **0.044**.